# ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

**General Certificate of Education Advanced Level**

**CHEMISTRY** **6031/1**

PAPER 1 Multiple Choice

**JUNE 2019 SESSION** 1 hour

Additional materials:
- Data Booklet
- Scientific/Electronic calculator
- Multiple Choice answer sheet
- Soft pencil (type B or HB is recommended)

**TIME** 1 hour

**INSTRUCTIONS TO CANDIDATES**

**Do not open this booklet until you are told to do so.**

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

**Read very carefully the instructions on the answer sheet.**

**INFORMATION FOR CANDIDATES**

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

**This question paper consists of 14 printed pages and 2 blank pages.**



**[Turn over**

## Section A

*For each question there are four possible answers, A, B, C and D. Choose the one you consider to be correct.*

**1** Which substance must be separated from chlorine by the diaphragm during the electrolysis of brine?

**A** hydrogen
**B** sodium chloride
**C** sodium hydroxide
**D** water

**2** Bromine reacts with hot concentrated alkali as shown in the equation.

$$3Br_{2(l)} + 6OH^-_{(aq)} \longrightarrow 5Br^-_{(aq)} + BrO_3^-{}_{(aq)} + 3H_2O_{(l)}$$

The oxidation state of bromine changes from zero to

**A** − 1 and +3.
**B** − 1 and +5.
**C** − 1 and +7.
**D** +1 and +7.

**3** Which equation correctly defines the standard enthalpy change of formation of hydrogen cynide, HCN?

**A** $H_{(g)} + C_{(s)} + N_{(g)} \longrightarrow HCN_{(g)}$

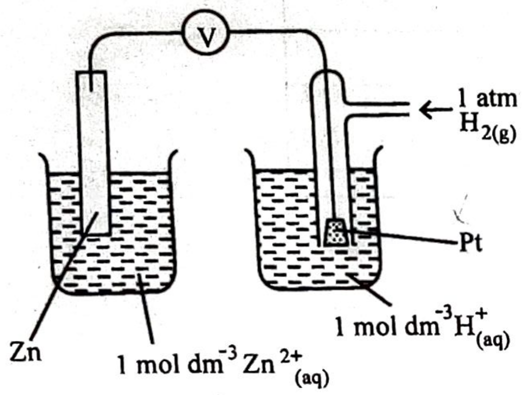
**B** $\frac{1}{2}H_{2(g)} + C_{(s)} + \frac{1}{2}N_{2(g)} \longrightarrow HCN_{(g)}$

**C** $H_{2(g)} + 2C_{(l)} + N_{2(g)} \longrightarrow 2HCN_{(g)}$

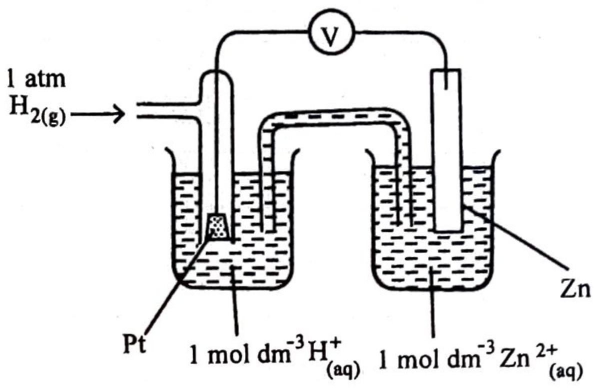
**D** $\frac{1}{2}H_{2(g)} + C_{(l)} + \frac{1}{2}N_{2(g)} \longrightarrow HCN_{(g)}$

4 Which diagram correctly shows how the standard hydrogen electrode is used to measure the $E^{\theta}$ of a zinc cell?

A




B

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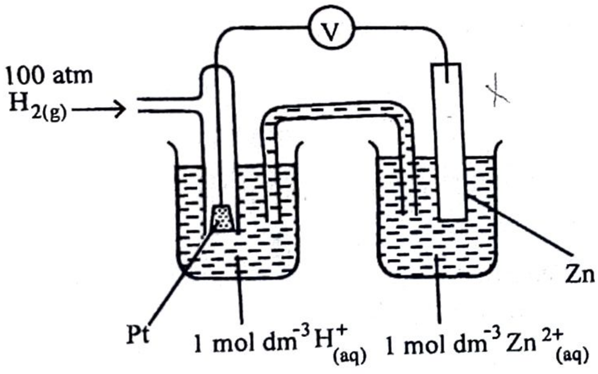
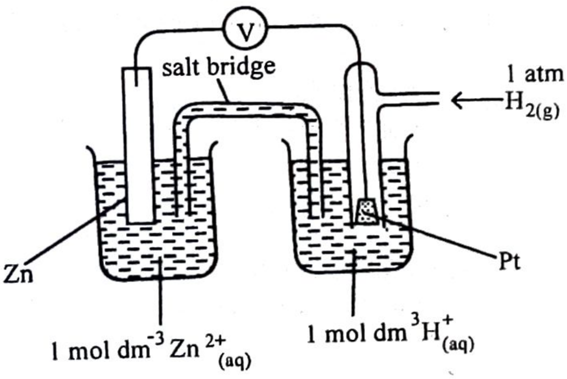



D




5 What are the anodic products when $NaBr_{(l)}$ and $CuF_{2(aq)}$ are electrolysed using inert electrodes?

A $Br_{2(g)}$ and $O_{2(g)}$
B $O_{2(g)}$ and $H_{2(g)}$
C $Br_{2(g)}$ and $F_{2(g)}$
D $Na_{(s)}$ and $F_{2(g)}$



[Turn over

6 The results shown were obtained for a reaction between **X** and **Y**.

| concentrations/$moldm^{-3}$ | | initial rate/$moldm^{-3}s^{-1}$ |
|---|---|---|
| [X] | [Y] | |
| 0.5 | 1.0 | 2 |
| 0.5 | 2.0 | 8 |
| 0.5 | 3.0 | 18 |
| 1.0 | 3.0 | 36 |
| 2.0 | 3.0 | 72 |

What is the rate equation for the reaction?

**A** k [Y]
**B** k[X] $[Y]^2$
**C** $k[X]^2[Y]$
**D** k[X][Y]

7 What are the units for the rate constant of a second-order reaction?

**A** $moldm^{-3}s^{-1}$
**B** $mol^{-1}dm^{3}s^{-1}$
**C** $mols^{-1}$
**D** $s^{-1}$

8 An excess of KI solution was added to a 25 $cm^3$ sample of a solution of a copper (II) salt. The liberated iodine required 15 $cm^3$ of 0.01 mol $dm^{-3}$ sodium thiosulphate to discharge its colour.

What is the concentration of $Cu^{2+}$ in the sample?

**A** 0.000006 mol $dm^{-3}$
**B** 0.00015 mol $dm^{-3}$
**C** 0.003 mol $dm^{-3}$
**D** 0.006 mol $dm^{-3}$

9 The reaction of acidified aqueous potassium iodide with aqueous hydrogen peroxide

$2I^-_{(aq)} + H_2O_{2(aq)} + 2H^+_{(aq)} \rightarrow I_{2(aq)} + 2H_2O_{(l)}$, is thought to involve the following steps:

$H_2O_2 + I^- \longrightarrow H_2O + OI^-$ (slow)
$OI^- + H^+ \longrightarrow HOI$ (fast)
$HOI + H^+ + I^- \longrightarrow I_2 + H_2O$ (fast)

Which statement about the reaction, is **not** true?

**A** The acid acts as a catalyst.
**B** The reaction is first order with respect to the iodide ion.
**C** The iodide ion is oxidised by the hydrogen peroxide.
**D** The overall order of the reaction is 2.

10 The value of the solubility product of $Ag_2CO_3$ is $2\times10^{-8}$.

What is the concentration of silver ions in $Ag_2CO_3$?

**A** $7.10 \times 10^{-5}$
**B** $1.00 \times 10^{-4}$
**C** $1.71 \times 10^{-3}$
**D** $3.42 \times 10^{-3}$

11 An element, X, forms both a dichloride $XCl_2$ and a tetrachloride $XCl_4$. Treatment of 10.00 g of $XCl_2$ with excess $Cl_{2(g)}$ forms 12.55 g of $XCl_4$.

What is element X?

**A** Pb
**B** Sn
**C** Ge
**D** Si

12 Berylium compounds are covalent due to

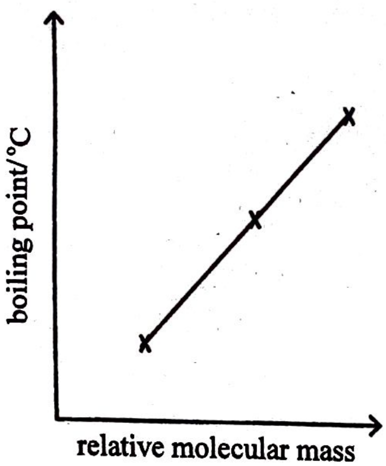
**A** low polarising power and large size of the berylium ion.
**B** low polarising power and small size of the berylium ion.
**C** high polarising power and large size of the berylium ion.
**D** high polarising power and small size of the berylium ion.

13 Which pair of oxides will give a solution with a very high pH?

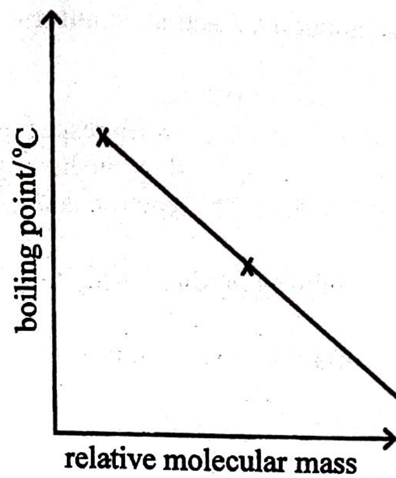
- **A** $Al_2O_3$ and MgO
- **B** $Na_2O$ and MgO
- **C** $Na_2O$ and $P_4O_{10}$
- **D** $SO_3$ and $P_4O_{10}$

14 Which graph correctly shows the variation of the boiling points of $SiCl_4$, $GeCl_4$ and $SnCl_4$?

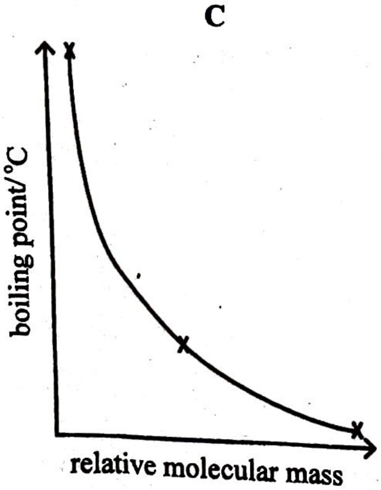
**A**



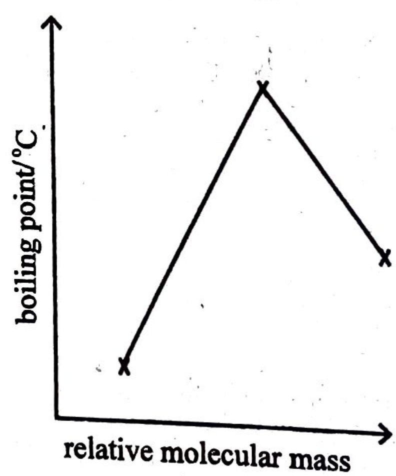

**B**




**C**




**D**



**15** **Which statement explains the difference in thermal stabilities of calcium nitrate and barium nitrate?**

**A** The lattice energy of calcium nitrate is smaller than that of barium nitrate.
**B** The ionic radius of the calcium ion is larger than that of the barium ion.
**C** The ionisation energy of calcium is greater than that of barium.
**D** The calcium ion has a greater charge density than that of barium.

**16** The pH of solution R was found to be 3 and the pH of another solution T was found to be 6.

It can be concluded that

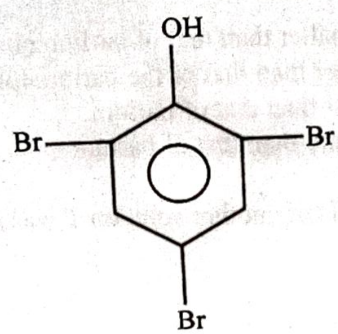
**A** $[H^+]$ in solution R is half that of $[H^+]$ in solution S.
**B** $[H^+]$ in solution R is double that of $[H^+]$ in solution S.
**C** $[H^+]$ in solution R is $1 \times 10^3$ times that of $[H^+]$ in solution S.
**D** $[H^+]$ in solution R is $1 \times 10^{-3}$ times that of $[H^+]$ in solution S.

**17** Which reagent could be used to distinguish between $CH_3CH(OH)CH_2CHO$ and $CH_3COCH_2CH_2OH$?

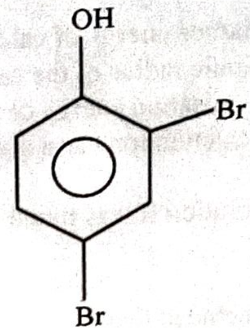
**A** 2,4-dinitrophenylhydrazine
**B** acidified potassium dichromate (VI)
**C** sodium carbonate
**D** Fehling's reagent

**18** Phenol reacts with excess bromine at room temperature to produce

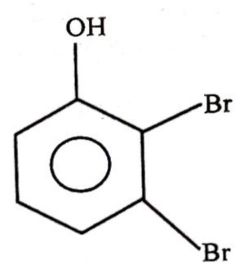
**A**



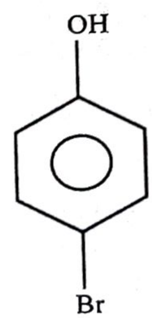
**B**



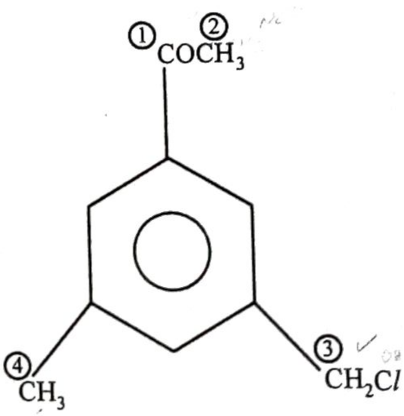
**C**



**D**



**19** The structure of an organic compound X is shown. The side chain carbon atoms are numbered 1 to 4.



On which carbon atom(s), 1, 2, 3, and/or 4, does nucleophilic addition attack occur?

**A** 1 and 2
**B** 2 and 3
**C** 1 and 3
**D** 2 and 4

**20** **Ethanal may be converted to a three carbon acid in a two step process.**

**Which compound is the intermediate?**

**A** $CH_3COCN$

**B** $CH_3CH_2CN$

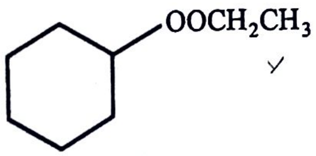
**C** $CH_3CH(OH)CN$

**D** $CH_2OHCH_2(OH)CN$

**21** **An organic compound, of molecular formula $C_6H_{12}O$, reacts as shown**

$$C_6H_{12}O \xrightarrow[H^+]{Cr_2O_7^{2-}} N \xrightarrow[H^+]{C_2H_5OH} W$$

**What is the formula of W?**

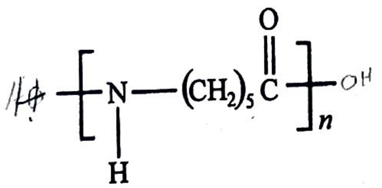
**A** 

**B** $CH_3(CH_2)_4CO_2CH_2CH_3$

**C** $CH_3CO_2H_2C(CH_2)_4CH_3$

**D** $CH_3(CH_2)_4COCH_2CH_3$

**22** **Nylon 6 has the formula shown and undergoes acid hydrolysis.**



**What is the product of the acid hydrolysis of nylon 6?**

**A** $HO^-(CH_2)_5CO_2H$

**B** $HO-(CH_2)_5OH$

**C** $H_2N-(CH_2)_5CO_2H$

**D** $H_2N^-(CH_2)_5OH$



**[Turn over**

23 Which one is the strongest acid?

A chloroethanoic acid

B ethanoic acid

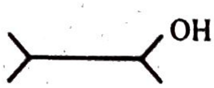
C ethylamine

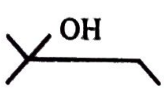
D phenol

24 Which reaction does not produce benzoic acid?

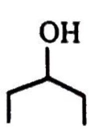
A hydrolysis of $C_6H_5CO_2CH_2CH_3$
B hydrolysis of $C_6H_5CN$
C oxidation of $C_6H_5CH_3$
D oxidation of $C_6H_5OH$

25 Which alcohol undergoes iodoform reaction?

A 

B 

C 

D 

26 Which one is **not** a benefit of carrying out a reaction at nano scale?

A fewer reagents are used
B increased productivity
C increased by-product(s)
D increased rate of reaction

27 Which process minimises the formation of acid rain?

A emitting photochemical oxidants into the atmosphere
B cutting down trees
C reducing discharge of industrial waste
D using lean-burn engines

28 In steam distillation, the organic liquid vaporises at

- **A** the boiling point of steam.
- **B** a temperature equal to its boiling point.
- **C** a temperature higher than its boiling point.
- **D** a temperature lower than its boiling point.

29 Which reagent reduces vanadium ion from +5 to +4 only?

$V_2O_5$

- **A** $SO_2$
- **B** $Sn^{2+}$
- **C** $Fe^{2+}$
- **D** Zn

30 Which statement correctly describes the properties of chromium and manganese?

- **A** both $Cr^{2+}$ and $Mn^{3+}$ are oxidising agents.
- **B** $Cr^{2+}$ is an oxidising agent while $Mn^{3+}$ is a reducing agent.
- **C** $Cr^{2+}$ and $Mn^{3+}$ have the same electronic configuration.
- **D** Only manganese has variable oxidation states.

Mg
C

## Section B

*For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.*

*Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).*

*The responses* **A** *to* **D** *should be selected on the basis of*

| A | B | C | D |
|---|---|---|---|
| *1, 2 and 3 are correct* | *1 and 2 only are correct* | *2 and 3 only are correct* | *1 only is correct* |

*No other combination of statements is used as a correct response.*

**31** Which complex ion(s) is/are paramagnetic?

1. $[VF_6]^-$
2. $[V(C_2O_4)_3]^{3-}$
3. $[V(CN)_6]^{4-}$

**32** Which statement(s) is/are true about $^{51}_{23}X^{3+}$, an ion of element X?

1. the electronic configuration of the ion is $1s^2 2s^2 2p^6 3s^2 3p^6$
2. X represents a transition metal
3. X forms ions of + 4 oxidation state

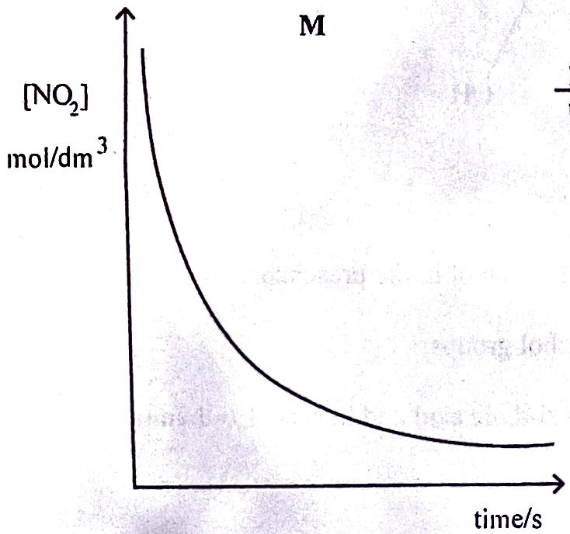
**33** The effect(s) of a catalyst on a reversible reaction, is/are to

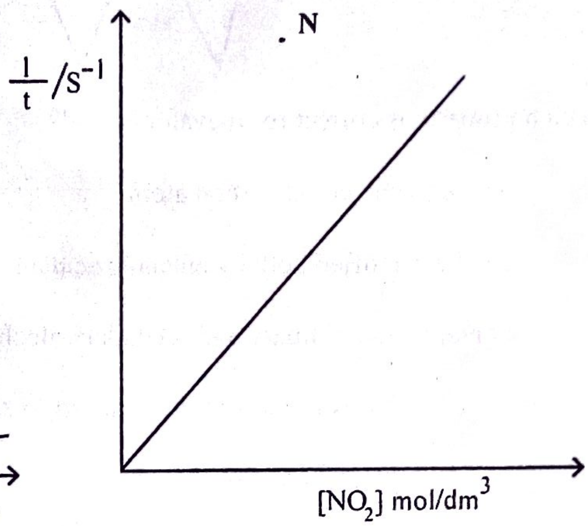
1. alter the mechanism of the reaction.
2. attain the equilibrium slowly.
3. attain a higher activation energy.

**34** Nitrogen dioxide decomposes according to the equation.

$$NO_{2(g)} \rightarrow NO_{(g)} + \frac{1}{2}O_{2(g)}$$

Graphs **M** and **N** were obtained when the reaction kinetics were investigated.




Which statements is/are true about the kinetics of the reaction?

1. the reaction is zero order with respect to $[NO_2]$
2. the rate constant can be determined from graph **N**
3. the initial rate can be determined from graph **M**

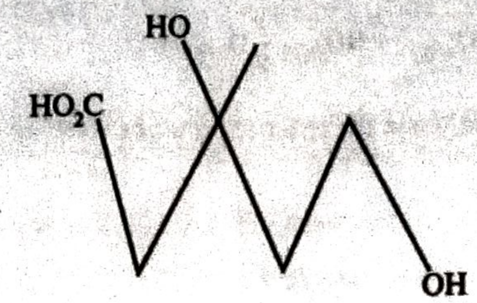
**35** Chlorine reacts with cold dilute alkali to form

1. $Cl^-$.
2. $ClO^-$.
3. $ClO_3^-$.

**36** Which statement(s) about magnesium sulphate is/are correct?

1. Its lattice energy is less than that of barium sulphate.
2. Its solubility in water is greater than that of barium sulphate.
3. It is added to acidic soils to neutralize excess acidity.

37 The structure of mevalonic acid is shown:



Which property is correct for mevalonic acid?

1. has only one chiral carbon atom
2. can be esterified both by ethanoic acid and ethanol in the presence of $H^+$ ions
3. contains both primary and secondary alcohol groups

38 During the formation of nylon-66 from hexane-1,6-dioic acid and hexane-1,6-diamine

1. amide linkages are formed
2. condensation polymerisation takes place
3. ammonia is eliminated

39 Which substance(s) evolve(s) ammonia gas when boiled with aqueous sodium hydroxide?

1. $CH_3CH_2NH_2$
2. $CH_3CONH_2$
3. $CH_3CONH_3^+$

40 Which statement(s) is/are true about gas liquid chromatograph (GLC) and high performance liquid chromatography (HPLC)?

1. GLC uses a longer column than HPLC.
2. In both GLC and HPLC, the stationery phase is a liquid.
3. In both GLC and HPLC, the mobile phase is a liquid.

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